Oral Microbiome 16S Analysis Using QIIME2

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Introduction

The human oral microbiome is a complex microbial ecosystem that contributes to oral health and disease. Shifts in community composition are linked to gingivitis, periodontitis, and other inflammatory conditions. 16S rRNA amplicon sequencing enables profiling of oral microbial communities by characterizing taxonomic composition, diversity, and between-sample differences. This analysis processes human oral paired-end 16S sequencing data using QIIME2 to generate ASVs, assign taxonomy, and evaluate alpha and beta diversity.

Objectives

- Organize paired-end FASTQ reads and prepare QIIME2-compatible manifest and metadata files.
- Perform quality inspection before and after primer trimming.
- Denoise and merge paired-end reads using DADA2 to generate ASVs.
- Assign taxonomy using a SILVA reference classifier.
- Filter non-target sequences (mitochondria and chloroplast) and construct a phylogenetic tree.
- Compute alpha and beta diversity metrics and test group differences using PERMANOVA.

Dataset Description

Sample type: Human oral microbiome
Samples analyzed: 4 paired-end samples
Sample IDs: SRR16600068, SRR16600076, SRR16600077, SRR16600081

Input files:

- Paired-end FASTQ files (stored in `raw_data/`)
- `manifest.csv` (QIIME2 manifest format)
- `metadata_qiime.tsv` (with `sample-id` as the first column)

Tools and Resources Used

- QIIME2 Docker container (amplicon pipeline)
- cutadapt (primer trimming)
- DADA2 (denoising and ASV inference)
- SILVA 138 Naive Bayes classifier (taxonomy assignment)
- QIIME2 View (visualization of `.qzv` outputs)

Methodology

FASTQ files were organized into a structured project directory and imported into QIIME2 using a manifest. Metadata was reformatted to meet QIIME2 requirements, including correcting the manifest header and ensuring the first metadata column was `sample-id`. Quality summaries were generated before and after primer trimming to assess read depth and per-base quality. Primer sequences were removed using cutadapt, with trimming settings evaluated to avoid unnecessary read loss. DADA2 was applied to filter low-quality reads, correct sequencing errors, merge paired-end reads, remove chimeras, and infer ASVs. Multiple denoising runs were performed to troubleshoot truncation and merging, focusing on read overlap and quality degradation. Outputs were summarized using feature table and denoising statistics visualizations. Taxonomy was assigned using a SILVA Naive Bayes classifier, with memory-related failures resolved by reducing compute load. The feature table was filtered to remove mitochondria and chloroplast, and representative sequences were synchronized with the filtered table to prevent feature ID mismatches. A phylogenetic tree was constructed using MAFFT alignment and FastTree, enabling phylogeny-based diversity analyses. Alpha rarefaction was used to select an appropriate sampling depth, followed by computation of alpha diversity (Shannon, Observed Features, Faith's PD) and beta diversity (Bray–Curtis, UniFrac, Jaccard). PERMANOVA was applied to test group-level differences.

Results Summary

Alpha Diversity

Shannon diversity values:

- SRR16600068: 6.2813
- SRR16600076: 6.6924
- SRR16600077: 6.3417
- SRR16600081: 6.0205

Observed ASVs:

- SRR16600068: 97
- SRR16600076: 127
- SRR16600077: 99
- SRR16600081: 90

Faith's Phylogenetic Diversity (PD):

- SRR16600068: 8.2580
- SRR16600076: 17.4583
- SRR16600077: 6.8012
- SRR16600081: 7.9926

SRR16600076 showed the highest phylogenetic diversity, while the remaining samples showed lower but comparable PD values.

Beta Diversity

Across Bray–Curtis and UniFrac-based Emperor plots, the four samples showed separation, indicating differences in microbial community composition and phylogenetic structure.

PERMANOVA (Healthy vs Periodontitis)

PERMANOVA did not show statistically significant differences between groups, which is expected given the small sample size (n = 4).

Key results:

- Weighted UniFrac: p = 0.319
- Bray–Curtis: p = 0.696
- Unweighted UniFrac: p = 0.315
- Jaccard: p = 0.671

Output Files Generated

- Read quality summaries before and after trimming
- Trimmed read artifacts
- DADA2 outputs: ASV table, representative sequences, denoising stats
- SILVA taxonomy classifications
- Filtered ASV table and representative sequences
- Rooted phylogenetic tree
- Rarefaction curves
- Core diversity metrics and PERMANOVA outputs

Key Issues Encountered

- Manifest header mismatch was corrected.
- Metadata format was corrected (`sample-id` requirement).
- Docker path confusion was resolved by using `/data` paths inside the container.
- Taxonomy classification required memory optimization.
- Feature ID mismatches occurred when outputs from different runs were mixed.
- Sampling depth errors were corrected by selecting depths consistent with the minimum sample frequency.

Figures and tables:

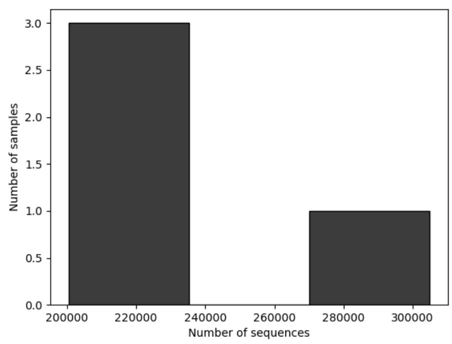
Visualization:

Read summary- QC reports- raw reads(raw, before primer trimming)

Demultiplexed sequence counts summary

	forward reads	reverse reads
Minimum	200402	200402
Median	202163.0	202163.0
Mean	227427.25	227427.25
Maximum	304981	304981
Total	909709	909709

Forward Reads Frequency Histogram



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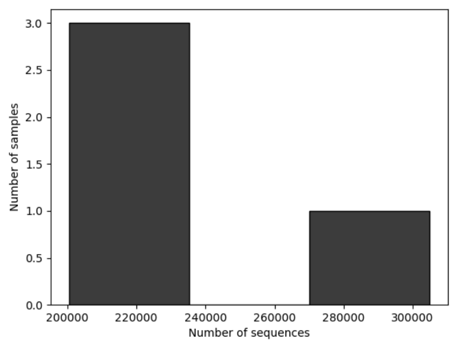
Reverse Reads Frequency Histogram



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Per-sample sequence counts

Total Samples: 4 (forward) 4 (reverse)

	forward sequence count	reverse sequence count
sample ID		
SRR16600077	304981	304981
SRR16600076	203606	203606
SRR16600068	200720	200720
SRR16600081	200402	200402

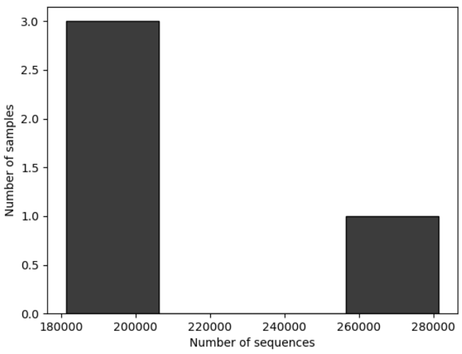
Download as TSV

QC report- Reads after primer trimming with cutadapt (with discard-untrimmed ON)

Demultiplexed sequence counts summary

	forward reads	reverse reads
Minimum	181220	181220
Median	187632.0	187632.0
Mean	209500.75	209500.75
Maximum	281519	281519
Total	838003	838003

Forward Reads Frequency Histogram



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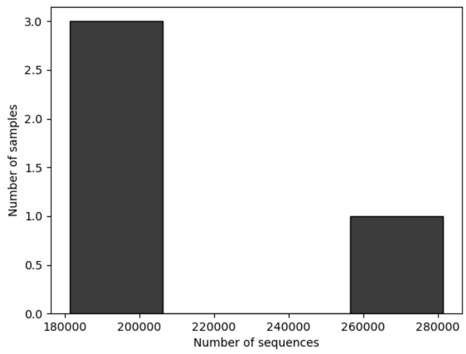
Reverse Reads Frequency Histogram



Download as PDF

Per-sample sequence counts

Total Samples: 4 (forward) 4 (reverse)

	forward sequence count	reverse sequence count
sample ID		
SRR16600077	281519	281519
SRR16600068	187699	187699
SRR16600081	187565	187565
SRR16600076	181220	181220

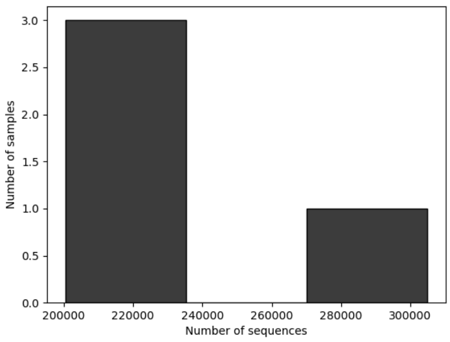
Download as TSV

QC report- Reads after primer trimming (without discarding untrimmed reads)

Demultiplexed sequence counts summary

	forward reads	reverse reads
Minimum	200402	200402
Median	202163.0	202163.0
Mean	227427.25	227427.25
Maximum	304981	304981
Total	909709	909709

Forward Reads Frequency Histogram



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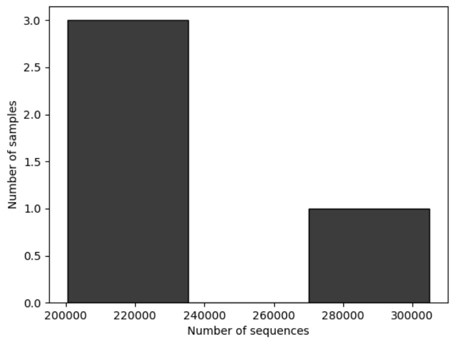
Reverse Reads Frequency Histogram



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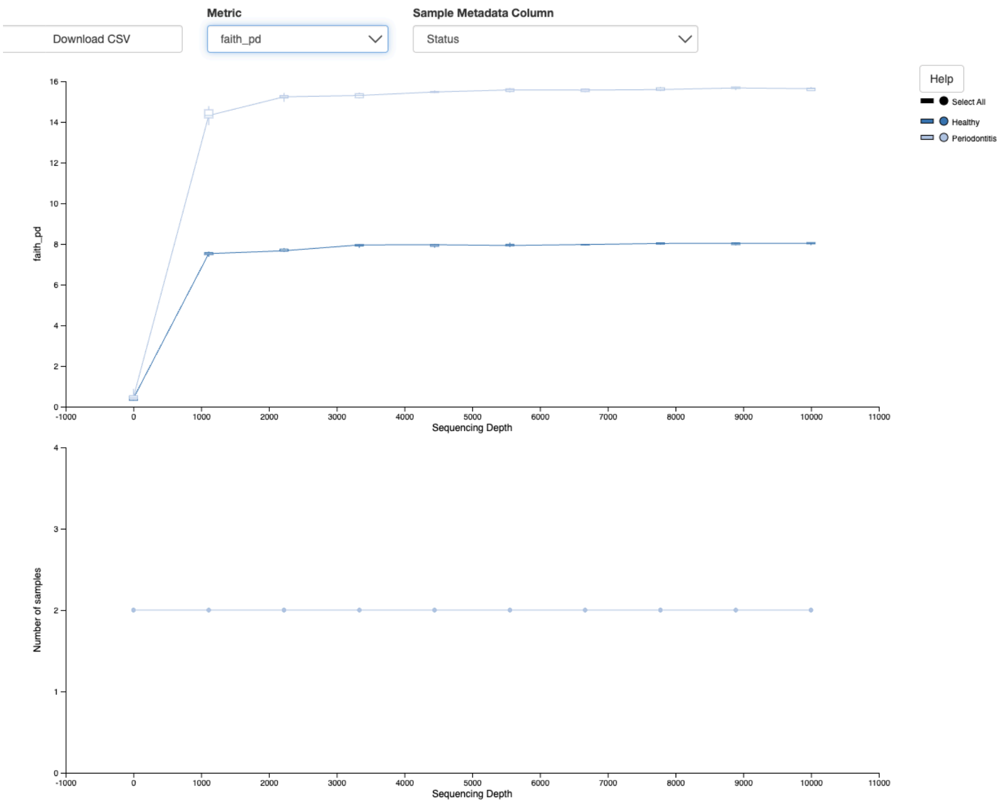
Per-sample sequence counts

Total Samples: 4 (forward) 4 (reverse)

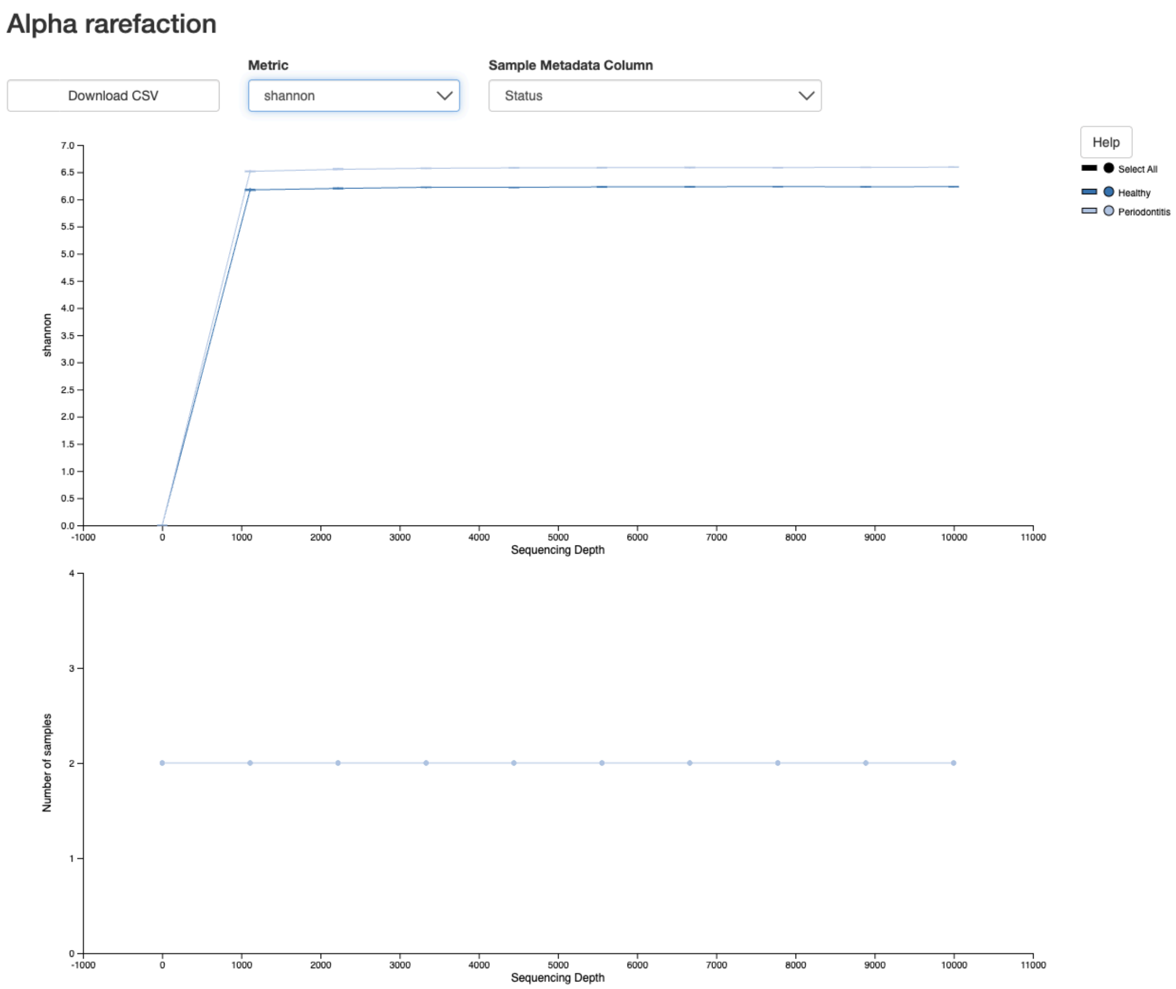
	forward sequence count	reverse sequence count
sample ID		
SRR16600077	304981	304981
SRR16600076	203606	203606
SRR16600068	200720	200720
SRR16600081	200402	200402

Download as TSV

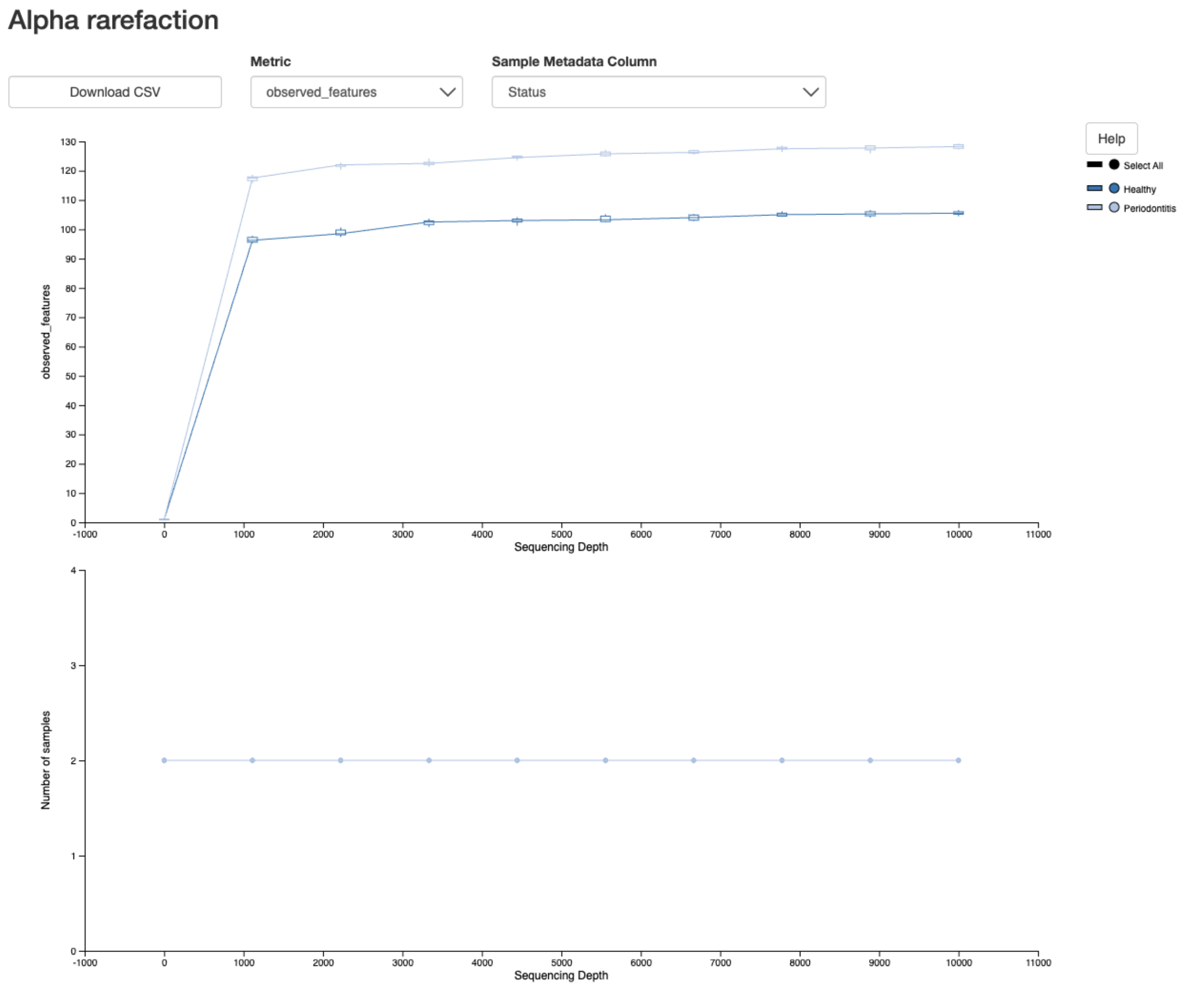
Rarefaction curves (alpha diversity vs sequencing depth). Used to choose sampling depth.



Faith PD (Faith's Phylogenetic Diversity):
Measures how much evolutionary tree branch length is covered by the sample. Higher = more phylogenetically diverse.



Shannon:
Measures richness + evenness (how many taxa + how evenly distributed). Higher = more diverse and balanced.



Observed Features:

Simple count of unique ASVs/features detected. Higher = more richness.

Summary after basic filtering (ex: low-frequency ASVs removed).

Original summary:

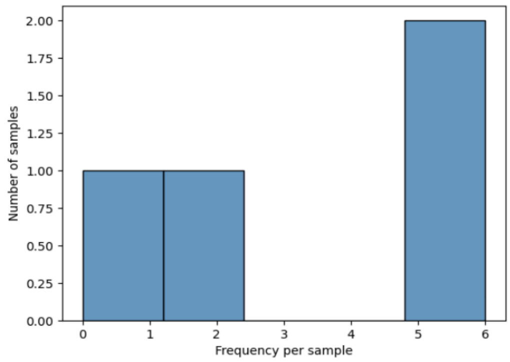
Table summary

Metric	Sample
Number of samples	4
Number of features	3
Total frequency	13

Frequency per sample

	Frequency
Minimum frequency	0.0
1st quartile	1.5
Median frequency	3.5
3rd quartile	5.25
Maximum frequency	6.0
Mean frequency	3.25

Frequency per sample detail (csv | html)



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Frequency per feature

	Frequency
Minimum frequency	2.0
1st quartile	3.5
Median frequency	5.0
3rd quartile	5.5
Maximum frequency	6.0
Mean frequency	4.333333333333333

Frequency per feature detail (csv | html)

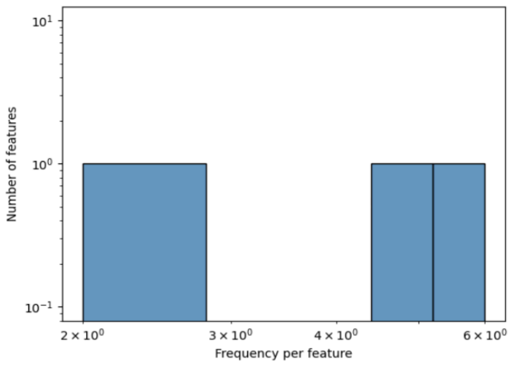


Table -filtering low frequency ASV

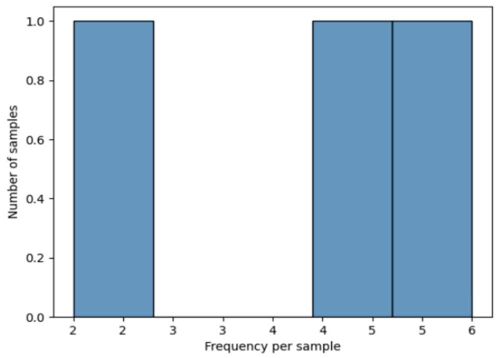
Table summary

Metric	Sample
Number of samples	3
Number of features	3
Total frequency	13

Frequency per sample

	Frequency
Minimum frequency	2.0
1st quartile	3.5
Median frequency	5.0
3rd quartile	5.5
Maximum frequency	6.0
Mean frequency	4.333333333333333

Frequency per sample detail (csv | html)



Download as PDF

Frequency per feature

	Frequency
Minimum frequency	2.0
1st quartile	3.5
Median frequency	5.0
3rd quartile	5.5
Maximum frequency	6.0
Mean frequency	4.333333333333333

Frequency per feature detail (csv | html)

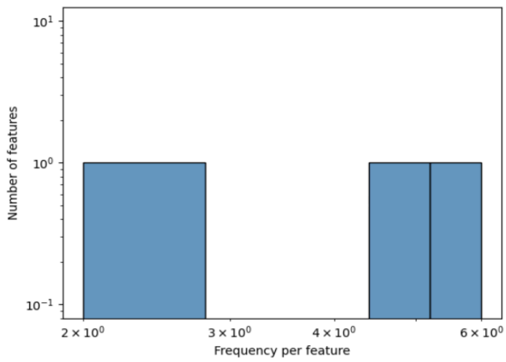


Table filtering contamination (mitochondria & chloroplast)

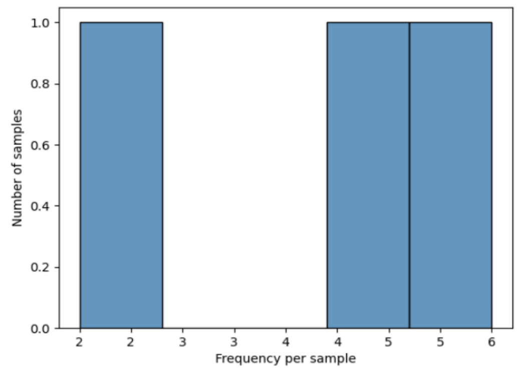
Table summary

Metric	Sample
Number of samples	3
Number of features	3
Total frequency	13

Frequency per sample

	Frequency
Minimum frequency	2.0
1st quartile	3.5
Median frequency	5.0
3rd quartile	5.5
Maximum frequency	6.0
Mean frequency	4.333333333333333

Frequency per sample detail (csv | html)



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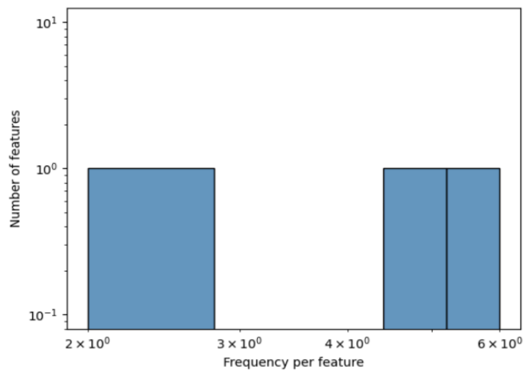
Frequency per feature

	Frequency
Minimum frequency	2.0
1st quartile	3.5
Median frequency	5.0
3rd quartile	5.5
Maximum frequency	6.0
Mean frequency	4.333333333333333

Frequency per feature detail (csv | html)



Alpha Diversity

Shannon Group significance:

Sample	Shannon entropy
SRR16600068	6.28129391523326
SRR16600076	6.6924287302841465
SRR16600077	6.3416537961402275
SRR16600081	6.020460599175123

Observed features vector:

Sample	Observed features
SRR16600068	97
SRR16600076	127
SRR16600077	99
SRR16600081	90

Faith phylogenetic diversity (PD)

Sample	Faith PD
SRR16600068	8.258039582808243
SRR16600076	17.4582823169895
SRR16600077	6.801248693956598
SRR16600081	7.99255253894972

SRR16600076 has the highest phylogenetic diversity (17.46), while the other three are lower and similar (~6.80–8.26)

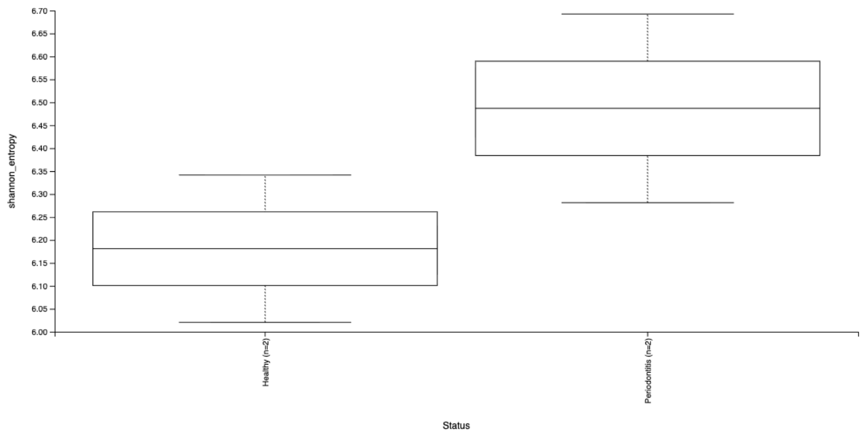
observed_features_group_significance

Alpha Diversity Boxplots

Download SVG

Column

Status



Download raw data as TSV

Kruskal-Wallis (all groups)

	Result
H	0.5999999999999996
p-value	0.43857802608099994

Kruskal-Wallis (pairwise)

Download CSV

Group 1	Group 2	H	p-value	q-value
Healthy (n=2)	**Periodontitis (n=2)**	0.6	0.438578	0.438578

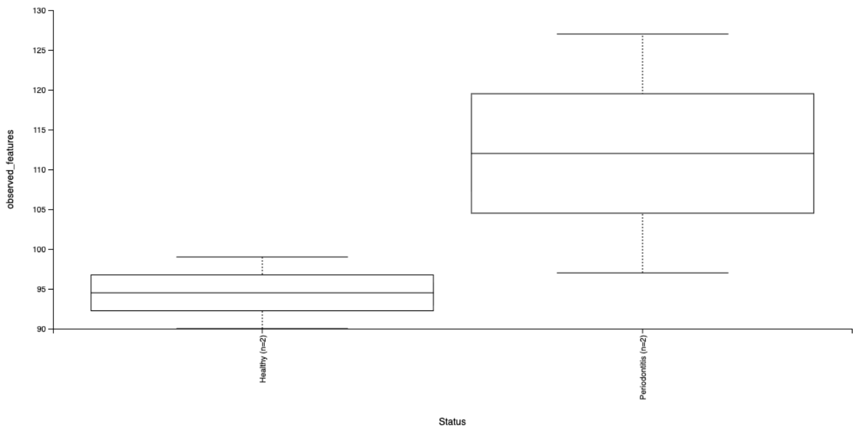
shannon_group_significance

Alpha Diversity Boxplots

Download SVG

Column

Status



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Kruskal-Wallis (all groups)

	Result
H	0.5999999999999996
p-value	0.43857802608099994

Kruskal-Wallis (pairwise)

Download CSV

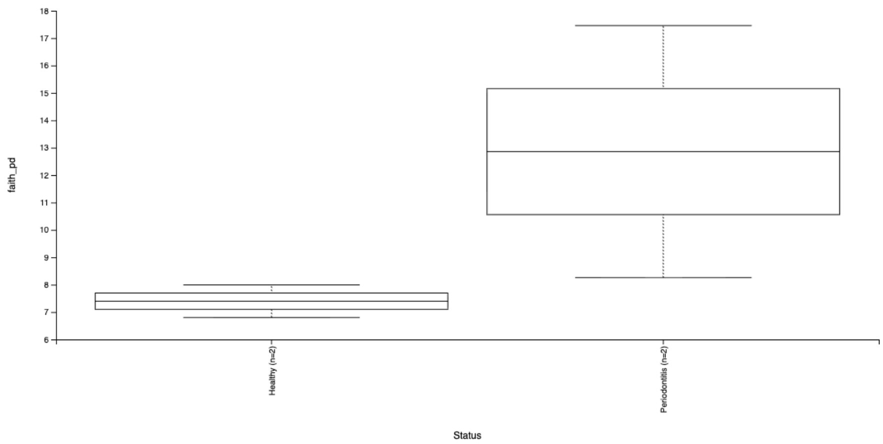
Group 1	Group 2	H	p-value	q-value
Healthy (n=2)	**Periodontitis (n=2)**	0.6	0.438578	0.438578

Faith Phylogenetic diversity:

Alpha Diversity Boxplots

Column

Download SVG | Status



Download raw data as TSV

Kruskal-Wallis (all groups)

	Result
H	2.3999999999999986
p-value	0.12133525035848367

Kruskal-Wallis (pairwise)

Download CSV

Group 1	Group 2	H	p-value	q-value
Healthy (n=2)	**Periodontitis (n=2)**	2.4	0.121335	0.121335

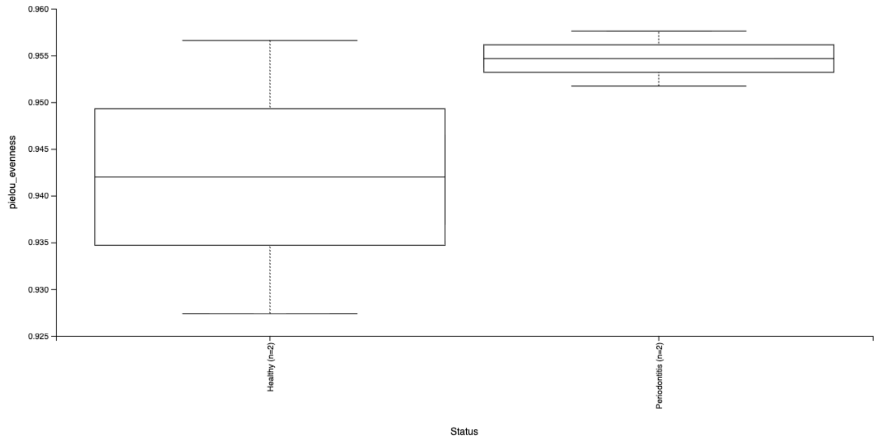
evenness_group_significance

Alpha Diversity Boxplots

Download SVG

Column

Status



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Kruskal-Wallis (all groups)

	Result
H	0.5999999999999996
p-value	0.43857802608099994

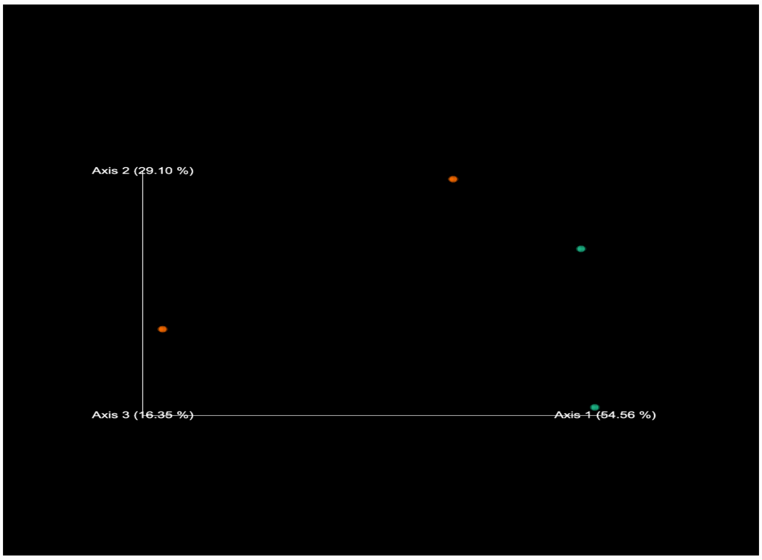
Kruskal-Wallis (pairwise)

Download CSV

Group 1	Group 2	H	p-value	q-value
Healthy (n=2)	**Periodontitis (n=2)**	0.6	0.438578	0.438578

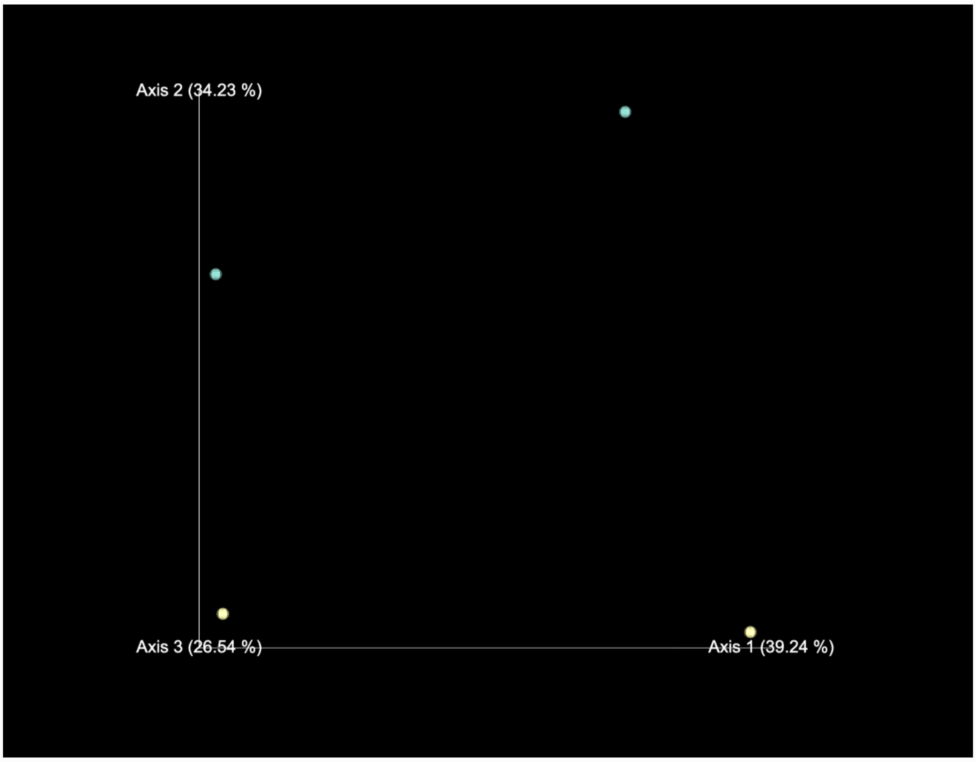
Beta diversity

weighted_unifrac_emperor



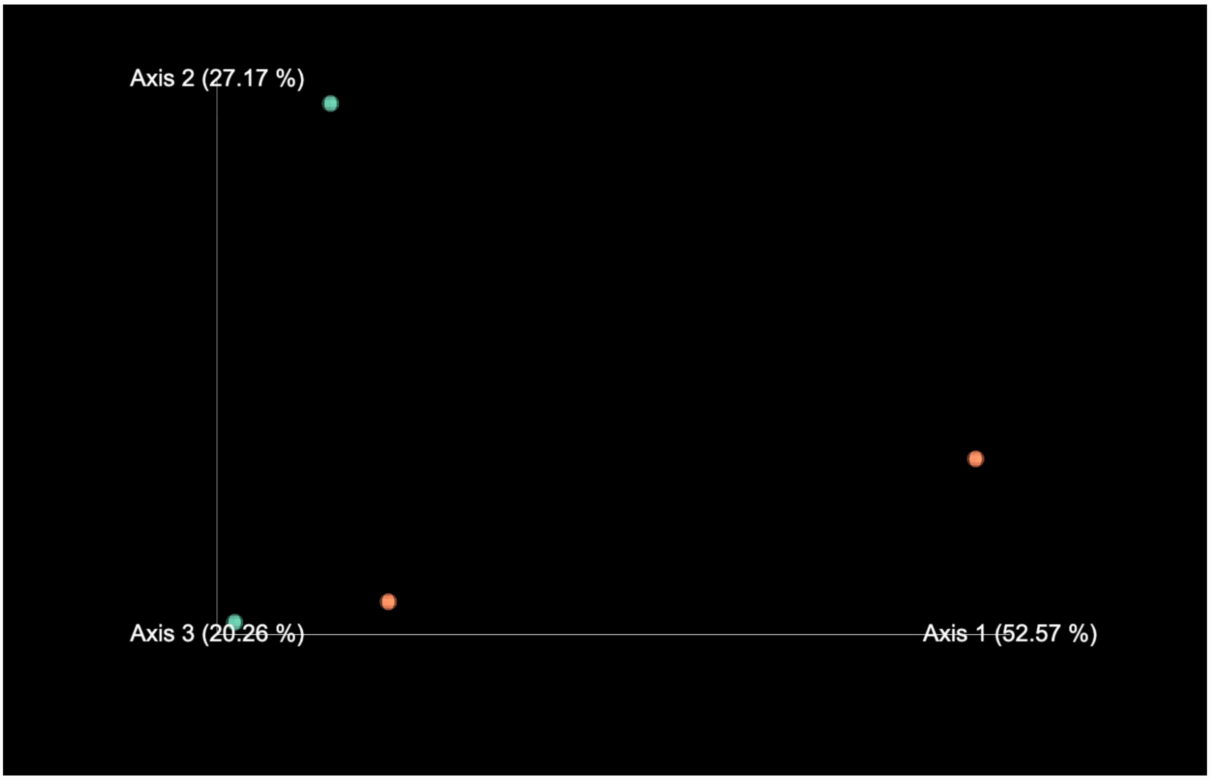
The weighted UniFrac Emperor plot shows the 4 samples are well separated, indicating clear differences in phylogenetic community structure considering relative abundance (not just presence/absence).

Bray curtis emperor:



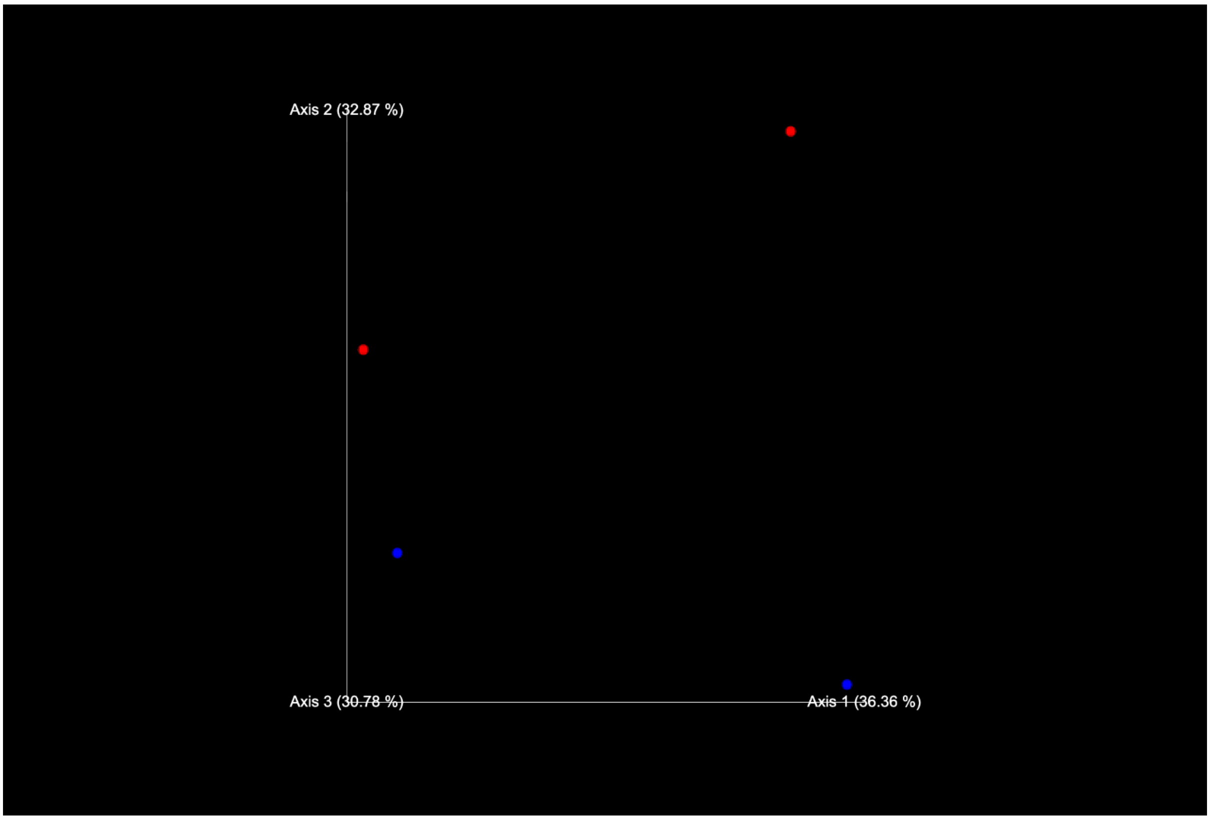
The Bray–Curtis Emperor plot shows the 4 samples are well separated, indicating clear differences in community composition between all samples (no tight clustering).

unweighted_unifrac_emperor



The unweighted UniFrac Emperor plot shows the samples are clearly separated, indicating strong differences in presence/absence of taxa (phylogenetic composition) between the 4 samples, with no clear clustering.

jaccard_emperor



The Jaccard Emperor plot shows the 4 samples are clearly separated, indicating strong differences in taxa presence/absence between samples, with little overlap in community membership.

Weighted unifrac status significance

qiime2view File: **weighted_unifrac_status_significance.qzv** × **Visualization** Citations Provenance Metadata

Overview

	PERMANOVA results
method name	PERMANOVA
test statistic name	pseudo-F
sample size	4
number of groups	2
test statistic	1.542834
p-value	0.318
number of permutations	999

Group significance plots

Download raw data as TSV

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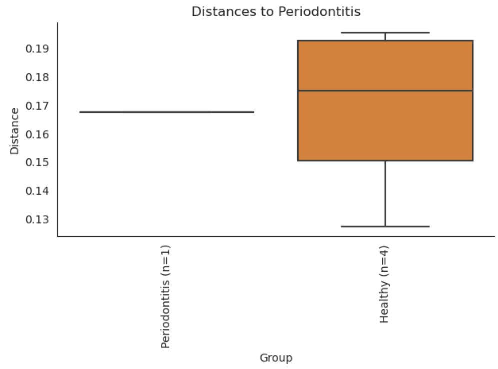


Download as PDF

Pairwise permanova results

Download CSV

Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
Healthy	**Periodontitis**	4	999	1.542834	0.319	0.319

Weighted UniFrac PERMANOVA (Healthy vs Periodontitis): there is no significant difference in community structure between groups (pseudo-F = 1.54, p = 0.319).
PERMANOVA interpretation (Healthy vs Periodontitis): the difference between groups is not statistically significant (pseudo-F = 1.54, p = 0.319, q = 0.319), likely limited by the very small sample size (n = 4)

bray_curtis_status_significance.

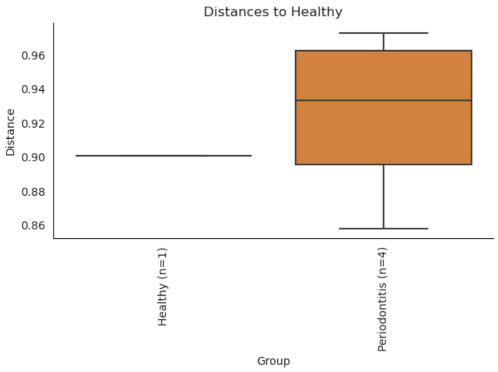
qiime2view   File: **bray_curtis_status_significance.qzv**   **Visualization**   Citations   Provenance   Metadata

Overview

	PERMANOVA results
method name	PERMANOVA
test statistic name	pseudo-F
sample size	4
number of groups	2
test statistic	1.028109
p-value	0.696
number of permutations	999

Group significance plots

Download raw data as TSV



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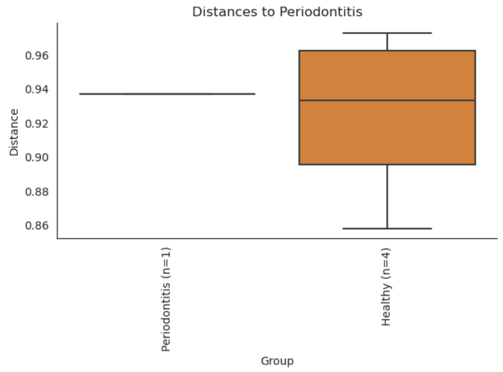


Download as PDF

Pairwise permanova results

Download CSV

Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
Healthy	**Periodontitis**	4	999	1.028109	0.683	0.683

Bray–Curtis PERMANOVA (Healthy vs Periodontitis): no significant difference in community composition between groups (pseudo-F = 1.03, p = 0.696; pairwise p = 0.683), likely affected by the very small sample size (n = 4).

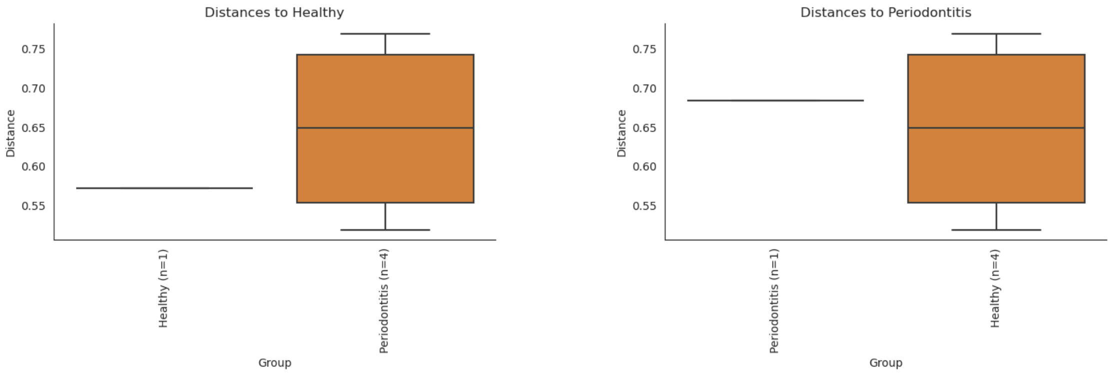
qiime2view   File: **unweighted_unifrac_status_significance.qzv**   **Visualization**   Citations   Provenance   Metadata

Overview

	PERMANOVA results
method name	PERMANOVA
test statistic name	pseudo-F
sample size	4
number of groups	2
test statistic	1.159166
p-value	0.315
number of permutations	999

Group significance plots

Download raw data as TSV



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Pairwise permanova results

Download CSV

Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
Healthy	**Periodontitis**	4	999	1.159166	0.327	0.327

Unweighted UniFrac PERMANOVA (Healthy vs Periodontitis): no significant difference in phylogenetic community membership (presence/absence) between groups (pseudo-F = 1.16, p = 0.315; pairwise p = 0.327, q = 0.327), likely limited by the small sample size (n = 4).

QIIME 2 Library View Forum

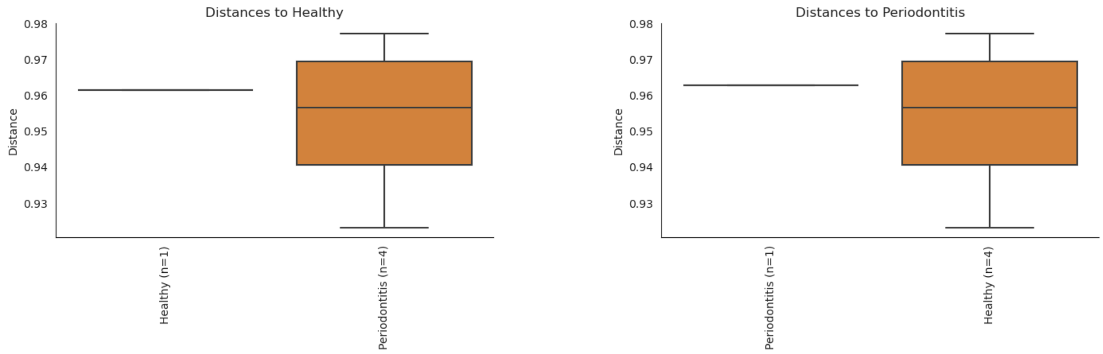
qiime2view File: jaccard_status_significance.qzv Visualization Citations Provenance Metadata

Overview

	PERMANOVA results
method name	PERMANOVA
test statistic name	pseudo-F
sample size	4
number of groups	2
test statistic	0.96462
p-value	0.671
number of permutations	999

Group significance plots

Download raw data as TSV



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Pairwise permanova results

Download CSV

Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
Healthy	Periodontitis	4	999	0.96462	0.658	0.658

Jaccard PERMANOVA (Healthy vs Periodontitis): no significant difference in taxa presence/absence between groups (pseudo-F = 0.965, p = 0.671; pairwise p = 0.658, q = 0.658), likely limited by the small sample size (n = 4).

Conclusion

This workflow successfully generated ASVs, taxonomy assignments, phylogenetic outputs, and diversity metrics for four oral microbiome 16S samples using QIIME2. While group-level statistical testing did not detect significant differences, this is primarily due to the limited sample size. The analysis provides a reproducible pipeline framework that can be scaled to larger datasets for robust comparisons of oral microbiome profiles in health and disease.